Section 6.1 Eigen Values and Eigen Vectors

Eigen is a German word meaning own. An eigen vector of a linear transformation is a special vector that intrinsically characterizes the linear transformation itself. In essence an Eigen vector is a vector for which its associated linear transformation A does not change under the action of A, except by a scaling.

Ex.

Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ What happens to the vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$?

eigen value

eigen vector

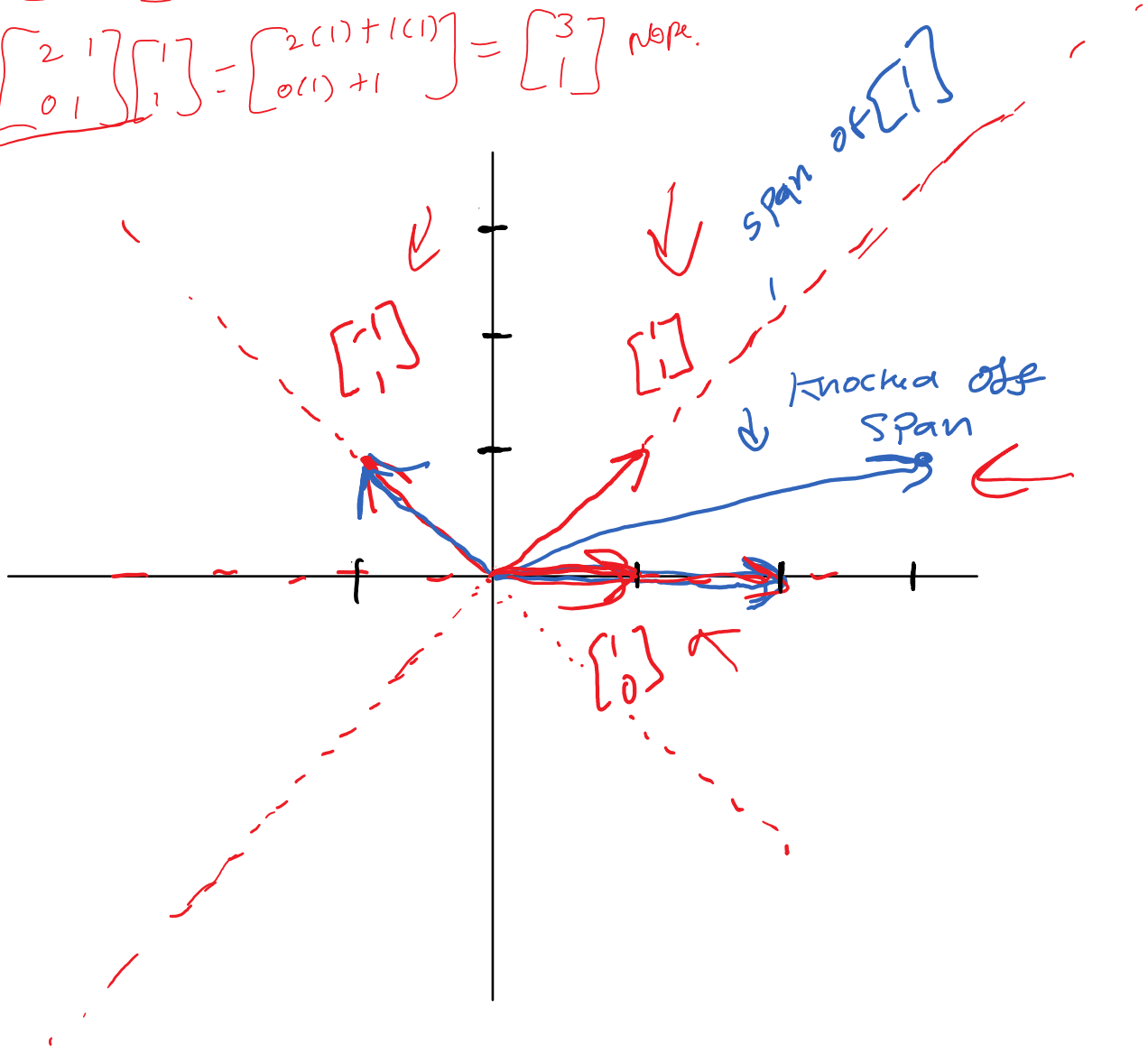
$\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2(1)+1(0) \\ 0(1)+1(0) \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

eigen value

eigen vec

$\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2(-1)+1(1) \\ 0(-1)+1(1) \end{bmatrix} = \begin{bmatrix} -2+1 \\ 1 \end{bmatrix} = 1\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2(1)+1(1) \\ 0(1)+1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ nope.



Section 6.1 Eigen Values and Eigen Vectors

For $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ It turns out that $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ are ***eigenvectors of A***

The vector $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ was scaled by 2 and the vector $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ was scaled by 1. These are called ***eigen values.***

Let A be an $n \times n$ matrix. Then a nonzero vector $\vec{u}$ is an ***eigenvector*** of A if there exists a scalar λ such that

$$A\vec{u} = \lambda\vec{u}$$

The scalar λ is called an ***eigenvalue*** of A

How do we find eigen values and vectors?

If we rearrange the equation in the definition

$$A\vec{u} = \lambda\vec{u} \rightarrow A\vec{u} - \lambda\vec{u} = 0 \rightarrow A\vec{u} - \lambda I\vec{u} = 0 \rightarrow (A - \lambda I)\vec{u} = 0$$

$$\left(\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \vdots \\ \vdots & 0 & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix} \right) \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = 0$$

$$\left(\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} - \begin{bmatrix} \lambda & 0 & \cdots & 0 \\ 0 & \lambda & 0 & \vdots \\ \vdots & 0 & \ddots & 0 \\ 0 & \cdots & 0 & \lambda \end{bmatrix} \right) \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = 0$$

$$\left(\begin{bmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{bmatrix} \right) \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = 0$$

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$$\left(\begin{bmatrix} a_{11}-\lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22}-\lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn}-\lambda \end{bmatrix}\right)\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = 0$$

So in looking for eigen values and eigen vectors we are searching values λ and ***non zero*** vectors $\vec{u}$ such that when we subtract λ from the diagonal entries of A and multiply that matrix by $\vec{u}$ we get zero. We can think of $\vec{u}$ as a vector in the null space of $(A-\lambda I)$.

Let $S=\{\vec{a}_1,\cdots,\vec{a}_n\}$ be a set of n vectors in R^n, let $A=[\vec{a}_1\ \vec{a}_2\ \cdots\ \vec{a}_n]$. Let $T: R^n \to R^n$ be given by $T(\vec{x})=A\vec{x}$. Then the following are equivlent:

(a) S *spans* R^n
(b) S *is linearly independent*
(c) $Ax=b$ *has a unique solution for all* $\vec{b}\in R^n$
(d) T *is onto*
(e) T *is one − to − one*
(f) A *is invertable*
(g) ker(T)=0 or equivalently the Nullspace(A)=0
(h) S *is a basis for* R^n
(i) $col(A)=R^n$
(j) $row(A)=R^n$
(k) $Rank(A)=n$
(l) **~~det(A) ≠ 0~~**

If not Nullspace(A)=0 then not $\det(A)\neq 0$ $= \det(A)=0$

What we want here is for $Nullspace(A-\lambda I)\neq \vec{0}$ we want nonzero vectors in the null space, those are the eigen vectors. From the unifying theorem above what does that say the $\det(A-\lambda I)$ must be?

$$\det(A-\lambda I)=0$$

This condition can be thought of geometrically as finding values of λ for which the matrix $(A-\lambda I)$ takes area to zero. That is collapses the plane onto the eigen vectors.

Section 6.1 Eigen Values and Eigen Vectors

So we want $\det(A - \lambda I) = 0$ this equation will allow us to find eigen values, then after we have those we can find the nullspaces of each $(A - \lambda I)$ to get eigenvectors.

Let's find the values for our previous example $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 2-\lambda & 1 \\ 0 & 1-\lambda \end{bmatrix}$ want $\begin{vmatrix} 2-\lambda & 1 \\ 0 & 1-\lambda \end{vmatrix} = 0$

$(2-\lambda)(1-\lambda) - 0(1) = 0$

$(2-\lambda)(1-\lambda) = 0 \rightarrow \lambda = 2, \lambda = 1$

We call the polynomial we get from $\det(A - \lambda I)$ the characteristic polynomial

To find our eigen vectors we will find the null spaces of

$\lambda = 2$

$(A - 2I) = \begin{bmatrix} 2-2 & 1 \\ 0 & 1-2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & -1 & 0 \end{array}\right] \rightarrow R_1 + R_2 \left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$

$0x_1 + 1x_2 = 0$

$0x_1 - 1x_2 = 0$

$x_2 = 0 \quad x_1 = t$

$-x_2 = 0$

$x_2 = 0$

$\begin{bmatrix} t \\ 0 \end{bmatrix} = t \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\lambda = 1$

$(A - 1I) = \begin{bmatrix} 2-1 & 1 \\ 0 & 1-1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$

$x_2 = t$

$x_1 + x_2 = 0$

$x_1 = -t$

$\begin{bmatrix} -t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$x_1 + x_2 = 0 \rightarrow x_1 = -x_2$

$0x_1 + 0x_2 = 0$

Section 6.1 Eigen Values and Eigen Vectors

Let's do a more complicated example together

Find the Eigen Values and Eigen Vectors of the matrix $A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix}$

$$\begin{vmatrix} 1-\lambda & 2 & 4 \\ 0 & -2-\lambda & 1 \\ 0 & 1 & -2-\lambda \end{vmatrix} = 0$$

Step 1. Find the characteristic polynomial an it's zeros will be the eigen values.

$$\det\left(\begin{bmatrix} 1-\lambda & 2 & 4 \\ 0 & -2-\lambda & 1 \\ 0 & 1 & -2-\lambda \end{bmatrix}\right) = 0$$

$$(1-\lambda)\begin{vmatrix} -2-\lambda & 1 \\ 1 & -2-\lambda \end{vmatrix} + 0\begin{vmatrix} 2 & 4 \\ 1 & -2-\lambda \end{vmatrix} + 0\begin{vmatrix} 2 & 4 \\ -2-\lambda & 1 \end{vmatrix} = 0$$

$$(1-\lambda)\big((-2-\lambda)(-2-\lambda)-1\big) = 0$$

$$(1-\lambda)\big(\lambda^2+4\lambda+4-1\big) = (1-\lambda)\big(\lambda^2+4\lambda+3\big) = (1-\lambda)(\lambda+1)(\lambda+3) = 0$$

$$\lambda_1 = 1 \quad \lambda_2 = -1 \quad \lambda_3 = -3$$

Step 2. Find the null spaces of the matrices $(A-\lambda_i I)$ those are your eigen vectors, each associated with an eigen value λ_i

$$(A-1I)\vec{x} = \begin{bmatrix} 0 & 2 & 4 \\ 0 & -3 & 1 \\ 0 & 1 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad (A-(-1)I)\vec{x} = \begin{bmatrix} 2 & 2 & 4 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad (A-(-3)I)\vec{x} = \begin{bmatrix} 4 & 2 & 4 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

```
octave:4> rref(A-eye(3))
ans =

   0   1   0
   0   0   1
   0   0   0
```

$x_1 = t$
$x_2 = 0$
$x_3 = 0$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t \\ 0 \\ 0 \end{bmatrix} = t\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\vec{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} t, \lambda_1 = 1$$

```
octave:5> rref(A-(-1)*eye(3))
ans =

   1   0   3
   0   1  -1
   0   0   0
```

$x_3 = t$
$x_2 - x_3 = 0 \rightarrow x_2 = t$
$x_1 + 3x_3 = 0 \rightarrow x_1 = -3t$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3t \\ t \\ t \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} t$$

$$\vec{u}_2 = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} t, \lambda_2 = -1$$

```
octave:8> rref(A-(-3)*eye(3))
ans =

   1.00000   0.00000   0.50000
   0.00000   1.00000   1.00000
   0.00000   0.00000   0.00000
```

$x_3 = t$
$x_2 + x_3 = 0 \rightarrow x_2 = -t$
$x_1 + \frac{1}{2}x_3 = 0 \rightarrow x_1 = -\frac{1}{2}t$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}t \\ -t \\ t \end{bmatrix}$$

$$\vec{u}_3 = \begin{bmatrix} -\frac{1}{2} \\ -1 \\ 1 \end{bmatrix} t, \lambda_3 = -3$$

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```
octave:11> u1=[1; 0; 0]
u1 =

   1
   0
   0
```

```
octave:12> u2=[-3; 1; 1]
u2 =

  -3
   1
   1
```

$A\vec{v}_2 = \lambda\vec{v}_2$

```
octave:13> u3=[-1/2; 1; 1]
u3 =

  -1/2
   1
   1
```

It works!

```
octave:19> A*u1
ans =

   1
   0
   0
```

$\lambda = 1$

$\begin{bmatrix}1\\0\\0\end{bmatrix} = 1\begin{bmatrix}1\\0\\0\end{bmatrix}$

```
octave:20> A*u2
ans =

   3
  -1
  -1
```

$\lambda = -1$

$\begin{bmatrix}3\\-1\\-1\end{bmatrix} = -1\begin{bmatrix}-3\\1\\1\end{bmatrix}$

```
octave:21> A*u3
ans =

   3/2
   3
  -3
```

$\lambda = -3$

$\begin{bmatrix}3/2\\3\\-3\end{bmatrix} = -3\begin{bmatrix}-1/2\\1\\1\end{bmatrix}$

So far we have seen examples when there are as many eigen values and vectors as column vectors, is that always how it is?

$$A = \begin{bmatrix}1 & 2 & 4\\ 0 & -2 & 1\\ 0 & 0 & -2\end{bmatrix}$$

$$\det(A - \lambda I) = \det\left(\begin{bmatrix}1-\lambda & 2 & 4\\ 0 & -2-\lambda & 1\\ 0 & 0 & -2-\lambda\end{bmatrix}\right) = (1-\lambda)(-2-\lambda)(-2-\lambda) = 0$$

$$\lambda_1 = 1, \lambda_2 = -2, \lambda_3 = -2$$

```
octave:29> (A-(1)*eye(3))
ans =

   0   2   4
   0  -3   1
   0   0  -3
```

$\lambda = 1$

$\begin{bmatrix}0 & 2 & 4 & 0\\ 0 & -3 & 1 & 0\\ 0 & 0 & -3 & 0\end{bmatrix} \rightarrow \begin{bmatrix}0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$

$x_1 = t$
$x_2 = 0$
$x_3 = 0$

$$\vec{u}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix} t, \lambda_1 = 1$$

```
octave:30> (A-(-2)*eye(3))
ans =

   3   2   4
   0   0   1
   0   0   0
```

$\lambda = -2$

```
octave:27> rref(A-1*eye(3))
ans =

   0   1   0
   0   0   1
   0   0   0
```

```
octave:28> rref(A-(-2)*eye(3))
ans =

   1  2/3   0
   0   0   1
   0   0   0
```

$\begin{bmatrix}1 & 2/3 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$

$x_3 = 0$
$x_2 = t$
$x_1 = -\frac{2}{3}t$

$$\vec{u}_2 = \begin{bmatrix}-\frac{2}{3}\\1\\0\end{bmatrix} t, \lambda_2 = -2$$

In this case we only get two eigen vectors! However repeated eigen values does not always preclude a loss of eigen vectors.

Section 6.1 Eigen Values and Eigen Vectors

```
octave:41> A=[1 -2 1; -1 0 1; -1 -2 3]
A =

   1  -2   1
  -1   0   1
  -1  -2   3
```

```
octave:42> eig(A)
ans =

   0
   2
   2
```

λ_1 λ_2 λ_3

$[A-0eye(3)] \to \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & -2 & 3 & 0 \end{array}\right]$

```
octave:44> rref(A-(0)*eye(3))
ans =

   1   0  -1   0
   0   1  -1   0
   0   0   0   0
```

```
octave:45> (A-(2)*eye(3))
ans =

  -1  -2   1
  -1  -2   1
  -1  -2   1
```

```
octave:45> (A-(2)*eye(3))
ans =

  -1  -2   1
  -1  -2   1
  -1  -2   1
```

```
octave:46> rref(A-(2)*eye(3))
ans =

   1   2  -1   0
   0   0   0   0
   0   0   0   0
```

one free paramater

$x_3 = t$

$x_2 - x_3 = 0 \quad x_2 - t = 0$

$x_2 = t$

$x_1 - x_3 = 0$

$x_1 - t = 0$

$x_3 = t$

$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t \\ t \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} t$

$= \text{Span}\left\{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right\} \quad \lambda = 0$

2 free paramaters

$x_3 = t \quad x_2 = s$

$x_1 + 2x_2 - x_3 = 0$

$x_1 + 2s - t = 0$

$x_1 = -2s + t$

$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2s+t \\ s \\ t \end{bmatrix} = \begin{bmatrix} -2s \\ s \\ 0 \end{bmatrix} + \begin{bmatrix} t \\ 0 \\ t \end{bmatrix}$

$= \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} s + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} t$

$\text{span}\left\{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right\} \quad \lambda = 2$

Section 6.1 Eigen Values and Eigen Vectors

The set of eigen vectors associated with an eigen value λ together with the zero vector forms a subspace.

Proof: (this is a quiz problem for Thursday) $\vec{v}_1, \vec{v}_2 \in S \Rightarrow \vec{v}_1 + \vec{v}_2 \in S$

For this I'll give hints

Let $\vec{v}_1, \vec{v}_2$ be eigenvecs of A assoc. with λ

Look at $\vec{v}_1 + \vec{v}_2$

1. Show the sum of eigen vectors associated with λ is still an eigen vector associated with λ
2. Show a scalar multiple of an eigen vector associated with lambda is still an eigen vector associated with λ.

$\vec{v}_1 \in S \quad r\vec{v}_1 \in S \quad r \in \mathbb{R}$

Let A be a square matrix with eigenvalue λ. The subspace of all eigenvectgors associated with λ, together with the zero vector is called the eigenspace of λ.

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

Eigen space $\lambda_1 = 1$

$$span\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right\}$$

Eigen space $\lambda_2 = -2$

$$span\left\{\begin{bmatrix} -\frac{2}{3} \\ 1 \\ 0 \end{bmatrix}\right\}$$ ← eigen space of $\lambda = -2$

$$A = \begin{bmatrix} 1 & -2 & 1 \\ -1 & 0 & 1 \\ -1 & -2 & 3 \end{bmatrix}$$

Eigen space $\lambda_1 = 0$

$$span\left\{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right\}$$

Eigen space $\lambda_2 = 2$

$$span\left\{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}\right\}$$

Section 6.1 Eigen Values and Eigen Vectors

In the previous two examples we had a repeated eigen value , 2 and -2,. Each showed up twice in the characteristic polynomial and so we would say it has multiplicity 2. One had eigenspace with dimension 1 and one with dimension 2.

Let λ be an eigen value of an $n \times n$ matrix A of multiplicity m. Then the dimension of the eigen space is always less than or equal to m.

The Unifying Theorem Expanded Again!

Let $S = \{\vec{a}_1, \cdots, \vec{a}_n\}$ be a set of n vectors in R^n, let $A = [\vec{a}_1 \ \vec{a}_2 \ \cdots \ \vec{a}_n]$. Let $T: R^n \to R^n$ be given by $T(\vec{x}) = A\vec{x}$. Then the following are equivlent:

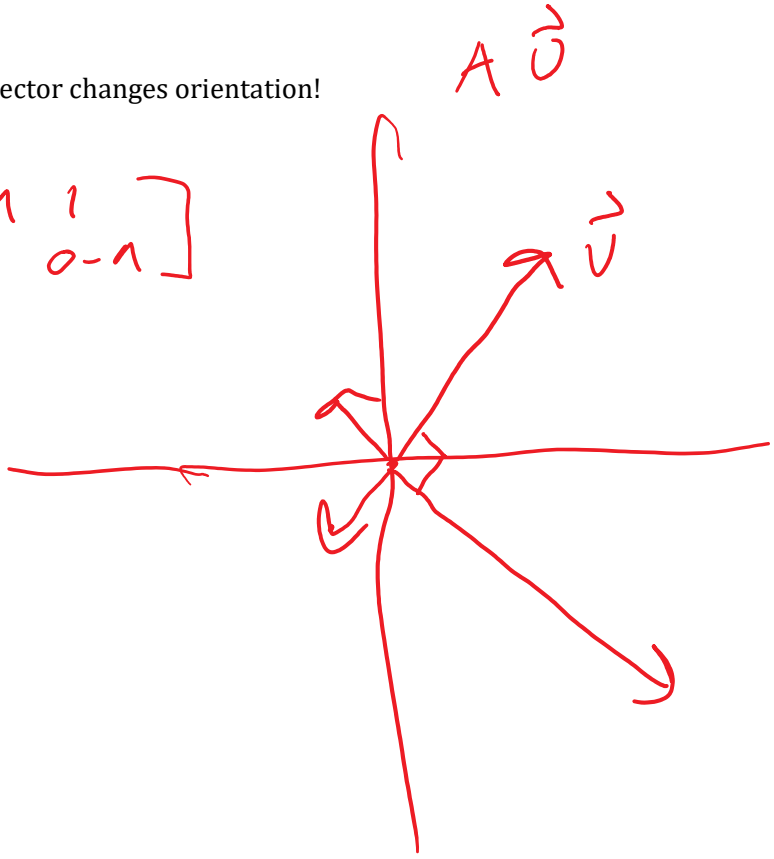
(a) S spans R^n
(b) S is linearly independent
(c) $Ax = b$ has a unique solution for all $\vec{b} \in R^n$
(d) T is onto
(e) T is one $-$ to $-$ one
(f) A is invertable
(g) ker(T)=0 or equivalently the Nullspace(A)=0
(h) S is a basis for R^n
(i) $col(A) = R^n$
(j) $row(A) = R^n$
(k) $Rank(A) = n$
(l) $\det(A) \neq 0$
$(\boldsymbol{m})$ $\boldsymbol{\lambda = 0}$ ***is not an eigen value of*** $\boldsymbol{A}$

Section 6.1 Eigen Values and Eigen Vectors

When there are no eigen vectors!

Think about a rotation in 2d, every vector changes orientation!

$$\begin{bmatrix} \cos\left(\frac{-\pi}{2}\right) & -\sin\left(\frac{-\pi}{2}\right) \\ \sin\left(\frac{-\pi}{2}\right) & \cos\left(\frac{-\pi}{2}\right) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$



$$\det\left(\begin{bmatrix} 0-\lambda & 1 \\ -1 & 0-\lambda \end{bmatrix}\right) = 0 \rightarrow \lambda^2 + 1 = 0, \lambda = \pm i$$

$$(0-\lambda)(0-\lambda) - (-1)(1)$$

$$\lambda^2 + 1 = 0$$

$$\lambda^2 = -1$$

$$\lambda = \pm i$$

In general we can have complex eigen values and complex eigenvectors, but not in the space of real numbers that we are sticking to in this class. However, complex eigenvalues of matrices say a lot about physical systems. If you take a differential equations class you will run into them there!

For the system of differential equations given by

$$\begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

$$\Rightarrow \frac{dx_1}{dt} = x_2$$

$$\Rightarrow \frac{dx_2}{dt} = -x_1$$

